

## Project Proposals

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### 1. Project Title

- Title: ReviewKita – AI Learning & Review Website

### 2. Project Description

#### Summary:

ReviewKita is a web app where students can upload notes, generate quizzes, and get AI-powered summaries. It transforms study materials into quick, personalized learning aids that help students review more effectively where it also includes achievements through gamified quizzes.

#### Scope:

##### Web Application:

- Upload users notes and highlight key sections.
- AI-generated quizzes and flashcards.
- Provide AI-powered summaries of uploaded materials.
- Offer a simple study progress tracker.

### 3. Objectives

1. Help students review faster with AI-generated quizzes and summaries.
2. Simplify study materials using LLM-powered summaries.
3. Provide a lightweight progress tracking system for monitoring learning outcomes.
4. Integration of gamified elements to boost motivation and engagement.
5. Enhance retention through gamified quizzes and achievements.

### 4. Target Users

1. **Students** – Review and test knowledge more effectively.

### 5. System Features

1. Secure user authentication and profile management.
2. Document upload (PDF/DOCX/Notes).

3. AI-generated quizzes, flashcards, and summaries.
4. Study progress dashboard and analytics.
5. Notifications & Reminders – quiz schedules and progress alerts.
6. Gamification Features – badges, streaks, leaderboards, and rewards for consistent study.

## 6. Technologies and tools

- **Frontend:**
  - Web: We selected [React.js](#) for its JavaScript foundation, ensuring our entire project remains within the JavaScript/[Node.js](#) ecosystem. React's comprehensive feature set enables us to develop responsive, modern user interfaces with strong visual appeal.
- **Backend:**
  - [Express.js](#): We choose [Express.js](#) as our backend framework due to its JavaScript/[Node.js](#) foundation, maintaining consistency across our full-stack development. [Express.js](#) offers extensive library support with the use of NPM, has a robust ecosystem integration, good for API development.
- **Database:**
  - **MongoDB**: We selected MongoDB as our database. MongoDB's NoSQL document structure offers superior flexibility and scalability compared to traditional SQL databases, with faster development cycles and horizontal scaling capabilities. For our LLM integration, MongoDB provides specialized Vector Search functionality and aggregation frameworks that efficiently handle unstructured AI-generated data.
- **Version Control:**
  - Github: For source code management and collaboration.

## 7. Designing Tools

- Figma: We chose **Figma** because our designer is already proficient with it, and its extensive plugin library helps speed up our prototyping workflow. This allows us to move quickly from design concepts to functional prototypes.